Airplane Flying Handbook (FAA-H-8083-3C)

Index

A

Abnormal engine instrument indication	**18-16**
Absence of propeller:	
Drag	**16-4**
Effect	**16-4**
Slipstream	**16-4**
Academic material (knowledge and risk management)	**5-5**
Prevention through ADM and risk management	**5-6**
Prevention through proportional counter-response	**5-6**
Recovery	**5-6**
Accelerated stalls	**5-19**
Accelerate-go distance	13-10, 13-12
Accelerate-stop distance	13-10, 13-12, 13-15, 16-12
After-landing	1-15, **2-22**
After-landing roll	2-22, 9-2, **9-9**, 9-10, 9-16, 9-18, 9-21, 9-22, 9-23, 9-36, 9-38, **14-6**, **14-7**, 14-8
Aircraft energy management	**4-18**
Airframe and systems	**17-5**
Airplane-based UPRT	**5-7**
Airplane configuration	3-21, 5-19, 6-1, 9-1, **18-4**
Airplane equipment and lighting	**11-5**
Airport and navigation lighting aids	**11-6**
Airport traffic patterns and operations	**8-1**
All-attitude/all-envelope flight training methods	**5-8**
All-engine service ceiling of multiengine airplanes	13-11
Alternator/generator	2-12, 13-6, **13-9,** 18-10
Angle of attack	1-13, 1-14, 4-5, 5-1, **5-9,** 5-10, 5-11, 5-12, 6-1, 9-8, 10-2, 12-1, 12-6, 13-19, 13-25, 13-26, 14-1, 15-8, 16-6, 16-7, 17-15
Anti-icing/deicing	**13-9**, 13-10
Approach (in an emergency)	**18-4**
Approach and landing (night)	**11-10**
Night emergencies	**11-13**
Approach and landing (sport)	**17-14**
Approaches to stalls (impending stalls), power-on, or power-off	**5-17**
Attitude and sink rate control	**18-4**
Attitude flying	**3-4**

Autoland systems	**18-22**

B

Ballistic parachute systems	**18-22**
Ballooning during round out	**9-32**
Bank control	3-5, 6-5, 18-18
Basic safety concepts	**18-2**
Before start and starting engine	**17-12**
Before-take-off check	**2-21**, 13-11, 13-15, 17-4
Avionics	2-22
Electrical system	2-21
Engine operation	2-21
Flight controls	2-21
Flight instruments	2-22
Fuel system	2-21
Takeoff briefing	2-22
Trim	2-21
Vacuum system	2-22
Bouncing during touchdown	**9-33**
Brakes	**2-10**, 2-11, 2-16, 2-17, 2-19, 2-20, 2-21, 2-22, 2-23, 6-3, 6-11, 9-9, 9-10. 9-18, 9-20, 9-23, 9-24, 9-36, 9-37, 9-38, 11-8, 13-14, 13-15, 13-16, 13-17, 13-21, 14-1, 14-2, 14-3, 14-7, 14-8, 16-9, 16-10, 16-12, 16-14, 16-15, 16-16, 16-19, 16-24, 17-12

C

Cabin fire	18-9, **18-10**
Captain's briefing	**16-13**, 16-14
Cascade reversers	16-11
Chandelle	**10-4**, 10-5, 10-6
Climb gradient	12-16, 13-2, 13-3, 13-10, 13-11, 13-12, 16-20
Climbs and climbing turns	**3-17**
Climbing turns	**3-20**
Establishing a climb	**3-19**
Best angle of climb (V_X)	3-17, 3-18, 4-17, 13-1, 16-17
Best rate of climb (V_Y)	3-17, 3-18, 4-17, 6-4, 6-6, 13-1, 13-16, 16-17
Normal climb	3-17, 6-12, 6-13, 11-8
Combustion heater	**13-8**
Constant radius turn during flight	**7-3**
Construction	**17-5**
Aluminum	17-5
Composite	17-5

Steel tube and fabric	17-5
Continuous ignition	15-1
Controllable-pitch propeller	3-19, 12-1, **12-5**, 12-6
Blade angle control	12-6, **12-7**
Climb	12-6, 12-7
Constant-speed propeller	10-5, **12-5**, 12-6, 12-7, 12-8, 13-3, 13-5, 13-9, 15-6, 18-9
Constant-speed propeller operation	**12-8**
Cruise	12-7
Fixed-pitch propellers	3-3, 10-5, 12-5, 12-8, 13-9
Governing range	**12-8**, 15-5
Takeoff	12-6
Control touch	1-1, 16-16
Coordinated flight	3-11, 3-12, 3-13, 3-23, 5-1, 5-6, **5-8**, 5-9, 5-11, 5-13, 5-16, 5-17, 5-18, 5-20, 6-5, 6-6, 6-14, 7-17, 9-3, 9-13, 10-5, 10-7, 13-21, 13-27
Coordination	1-1
Correcting drift during straight-and-level flight	**7-2**
Cross-control stall	**5-20**
Crosswind after-landing roll (tailwheel)	**14-7**
Crosswind approach and landing	9-3, 9-10, **9-15**, 9-17, **13-21**, 17-4
Crosswind after landing roll	**9-18**
Crosswind final approach	**9-15**
Crab method	9-15
Wing-low (sideslip) method	9-15, 9-16
Crosswind round out (flare)	**9-17**
Crosswind touchdown	**9-17**
Maximum safe crosswind velocities	**9-18**
Crosswind takeoff	**6-6, 13-15, 14-3**
Initial climb	**6-8**, 13-15, 13-16
Lift-off	**6-7**
Takeoff roll	**6-6**
Cruise	**17-13**

D

Defining an airplane upset	**5-1**
Descents and descending turns	**3-20**, 3-24, 5-10
Descent at minimum safe airspeed	3-21
Emergency descent	3-21, 10-3, 15-15, **18-8**, 18-9, 18-10, 18-22
Partial power descent	3-21
Detonation	2-7, 2-8, 12-8, 12-11

Directional control — 2-19, 3-9, 5-11, 5-17, 5-22, 6-1, 6-14, 9-9, 9-18, 9-23, 9-33, 9-34, 9-37, 9-38, 13-2, 13-15, 13-17, 13-18, 13-23, 13-25, 13-26, **13-29**, 13-30, 13-31, 13-32, 13-33, 13-36, 13-37, 14-1, 14-2, **14-3,** 14-5, 14-6, 14-7, 16-11, 16-14, 16-16, 16-24, 18-12, 18-13

Door opening in-flight — **18-17**

Drag devices — 4-4, **16-9**, 16-11

Drift and ground track control — **7-2**

E

Effect and use of flight controls — **3-1**

- Feel of the airplane — **3-3**

Electrical fires — 18-9

Elementary eights — **7-10**, 7-13,

- Eights across a road — 7-10, **7-12**
- Eights along a road — 7-10, **7-11**, 7-12
- Eights around pylons — 7-10, **7-13**, 7-14
- Eights-on-pylons — **7-14**, 7-15, 7-16, 7-17, 7-18

Elevator trim stall — **5-20**, 5-21

Emergencies — 3-23, 9-29, 11-5, 11-13, 13-35, 13-36, 15-13, 15-14, 15-15, **17-14**,

Emergency approaches and landings (simulated) — **9-28**

Emergency descents — 3-21, 10-3, 15-15, **18-8,** 18-9, 18-10, 18-22

Emergency landings — 9-28, **18-1**

- Psychological hazards — **18-1**
- Types of emergency landings — **18-1**
  - Ditching — 3-23, 18-1, **18-7**
  - Forced landing — 7-2, 9-12, 11-13, 12-6
  - Precautionary landing — 18-1, 18-8

Emergency situations — 11-1, 17-14, 18-1, 18-22, 18-23

Energy balance equation — 4-3, 4-4, 4-18

Energy error — 4-12, 4-14, 4-19

- Energy distribution error — 4-12, 4-13, 4-19
- Total energy error — 4-12, 4-19

Energy exchange — 4-2, 4-4, 4-5, 4-6, 4-8, 4-10, 4-18

Energy height or total specific energy (E_S) — 4-7, 4-9, 4-19

Energy state — 4-1, 4-2, 4-3, 4-6, 4-7, 4-8, 4-9, 4-10, 4-11, 4-12, 4-14, 4-18, 4-19, 5-13, 16-18

Energy system — 4-1, 4-18

Engine and propeller — **2-11**, 3-3, 6-1, 6-3, 13-12, 13-34, 15-3, 15-6

Engine failure:

- After lift-off — 13-11
- After takeoff — 13-31, 13-32, 13-33

After takeoff (single-engine)	**18-7**
During flight	**13-34**
Engine fire	13-32, 18-8, **18-9**
Engine inoperative approach and landing	**13-34**
Engine inoperative flight principles	**13-23**
Engines	**17-6**
Engine shutdown	**2-23**
Engine starting	**2-16**
Environmental factors	2-13, **5-2**
Exhaust gas temperature (EGT)	12-10, 12-11, 13-15

F

False start	15-12
Faulty approaches and landings	**9-30**
Feathering	2-21, 9-25, 13-3, 13-5, 13-6, 13-7, 13-29, 13-30, 13-35, 13-36, 15-1, 15-5, 18-9
Flap effectiveness	**12-3**
Flap control malfunction/failure	**18-10**
Asymmetric (split) flap	**18-11**
Landing gear malfunction	**18-12**
Loss of elevator control	**18-11**
Total flap failure	**18-10**
Flight director/autopilot	**13-8**
Flight environment	2-1, 15-13, 15-14, **17-8**
Flight standards service	**1-5**
Floating during round out	**9-32**
Forward slip	9-4, 9-12, 9-13, 9-14, **9-15**, 9-26, 9-30
Four fundamentals	**3-1**, 7-1
Climbs	**3-17**
Descents	**3-20**
Straight-and-level	**3-6**
Turns	**3-11**
Fowler flap	12-4
Fuel and oil	**2-7**
Fuel crossfeed	**13-7**
Full stalls:	
Power-off	**5-17**
Power-on	**5-18**
Function of flaps	**12-1**
Fundamentals of stall recovery	**5-15**

G

Gas turbine engine	**15-1**, 15-2, 15-8, 16-2
Glides	**3-21**, 3-23, 9-24
Gliding turns	**3-23**, 3-24
Go-around	**4**-17, 5-9, 5-17, 5-20, 5-21, 8-4, 9-6, **9-10**, 9-11, 9-12, 9-22, 9-30, 9-31, 9-32, 9-33, 9-34, 9-35, 12-4, 12-5, **13-22**, 13-23, 13-30, 13-33, 13-35, 14-5, 14-6, 16-4, 16-20, 16-22, 17-4
Rejected landings	**9-10**, 9-12
Attitude	**9-11**
Configuration	**9-11**
Ground effect	**9-12**
Power	**9-11**
Ground loop	2-19, 9-9, **9-36**, 9-37, 14-6, **14-8**, 18-5
Ground operation	2-1, 2-12, **2-14**, **13-14**, 14-1, 14-8, 15-3, 15-6, 17-4

H

Hand propping	**2-16**, 2-17
Hard landing	9-20, 9-22, 9-23, **9-35**, 12-4, 13-22, 16-23, 17-15
High final approach	**9-30**
High-performance airplane	12-1, 12-17, 12-18
High round out	9-10, **9-31**
Human factors	**5-2**
Diversion of attention	**5-2**
IMC	**5-2**
Sensory overload/deprivation	**5-3**
Spatial disorientation	**5-3**
Startle response	**5-3**
Surprise response	**5-3**
Task saturation	**5-3**
VMC to IMC	**5-2**
Hydraulic pump	12-12, 12-14, 13-30, 18-9
Hydroplaning	**9-37**, 9-38
Dynamic hydroplaning	**9-37**
Reverted rubber hydroplaning	**9-37**
Viscous hydroplaning	**9-37**

I

Impossible turn	18-7
Inadvertent flight into IMC	5-1, 5-2, 5-3, 5-26, **18-17**
Attitude control	3-5, 18-17, **18-18**

Climbs	**18-20**
Combined maneuvers	**18-21**
Descents	**18-20**
Maintaining airplane control	**18-18**
Recognition	**18-17**
Transition to visual flight	**18-22**
Turns	**18-19**
In-flight fire	15-14, **18-9**
Initial climb	6-1, **6-5**, 6-6, **6-8,** 6-10, 6-11, **6-12**, **6-13**, 6-14, 12-16, 13-15, 13-16, 13-17, 13-29, 14-4, 15-2, 15-15, 16-2, **16-17**,
Inside the airplane	2-21, 6-12, **17-9**
Instrumentation	3-3, 3-5, 5-2, 5-4, 11-4, **17-7**, 18-15, 18-23
Integrated flight instruction	**3-5**, 3-6
Intentional slips	**9-12**
Intentional spins	**5-25**, 5-26
Interstage turbine temperature (ITT)	15-4, 15-5, 15-8
Irreversible deceleration and/or sink rate	4-1, 4-5, 4-11, **4-14**, 4-17, 4-19

J

Jet airplane approach and landing	16-17, 16-18, 16-19, 16-20, 16-21, 16-22, 16-23, 16-24
Approach speed	**16-22**
Glidepath control	**16-22**
Landing requirements	16-19
Landing speeds	**16-20**
Approach climb	16-20
Landing climb	16-20
V_{REF}	16-20
V_{SO}	16-20
Stabilized approach	**16-21**
The flare	**16-23**
Touchdown and rollout	**16-24**
Jet engine basics	**16-1**
Jet engine efficiency	**16-4**
Jet engine ignition	16-1

K

Kinetic energy	4-1, 4-7, 4-9, 4-18, 4-19, 16-10, 16-18, 16-19

L

Landing	**9-2**

Landing gear	**2-10**, **14-1**
Instability	**14-1**
Landing gear control selected up, single-engine climb performance adequate	**13-31**
Checklist	**13-32**
Climb	1**3-32**
Configuration	**13-31**
Control	**13-31**
Landing gear selected up, singe-engine climb performance inadequate	13-30
Landing gear down	2-5, 13-16, **13-30**
Late or rapid round out	**9-31**
Lazy eight	**10-6**, 10-7
Level-off and cruise	3-20, 12-8
Level turns	**3-11**
Establishing a turn	**3-15**
Medium turns	3-12
Shallow turns	3-12
Steep turns	3-12
Turn radius	**3-14**
Liftoff	**6-4**
Rotation	6-5
Light sport airplane (LSA) background	**17-1**
Loss of control in-flight (LOC-I)	4-1, 4-19, 5-1, 5-2, 5-3, 5-4, 5-5, 5-6, 5-27
Low final approach	**9-30**
Low speed flight	12-1, **16-6**
LSA maintenance	17-4, **17-5**
LSA synopsis	**17-3**

M

Mach buffet boundaries	16-6
Manuevering by reference to ground objects	**7-1**
Minimum equipment list and configuration deviation list	16-24, **16-25**
Multiengine training considerations	**13-35**

N

Night illusions	**11-4**
Black-hole approach	11-5
Visual autokinesis	**11-4**
Night vision	**11-1**
Noise abatement	**16-14**

Normal and crosswind takeoff and climb	**13-15**, 17-4
Normal approach and landing	**9-2**, **13-20**
After-landing roll	**9-9**
Base leg	**9-2**
Final approach	**9-3**
Estimating height and movement	**9-7**
Use of flaps	**9-1**
Round out (flare)	**9-8**
Stabilized approach concept	**9-4**
Touchdown	**9-9**
Wrong surface landing avoidance	**9-4**
Normal takeoff	**6-3**
Intitial climb	**6-5**
Lift-off	**6-4**
Takeoff roll	**6-3**
Nose baggage compartment	**13-9**

O

Operating the jet engine	**16-2**
Operational considerations	**15-11**
Operation of systems	17-4, **13-3**
Orientation and navigation	**11-9**
Outer wing surfaces	**2-6**
Outside of the airplane	2-21, 18-18

P

Parking	**2-18**, 7-23, 11-8, 16-1, 17-4
Performance and limitations	**13-10**, 17-4
Pilot equipment	**11-5**
Pilot sensations in jet flying	**16-11**
Pitch and power	9-2, 9-3, 9-4, 9-20, 11-10, 12-4, 17-14
Pitch contol	3-2, 3-5, 3-17, 3-23, 3-24, 5-10, 5-21, 10-2, 10-3, 10-4, 13-32, 16-12, 16-24, 18-9, 18-11, 18-18
Plain (hinge) flap	12-3
Porpoising	**9-34**, 9-35, 18-13
Post-flight	**2-23**, **17-15**
Securing and servicing	2-23
Potential energy	4-1, 4-10, 4-18, 4-19, 16-18, 16-19
Power available	4-5, 4-14, 4-15, 4-16, 4-17, 4-18, 4-19, 13-3, 13-19, 13-29, 13-34, 16-20, 18-4
Power control	3-5, 4-6, 12-10, 16-2

Power-off accuracy approaches	**9-24**
90° Power-off approach	**9-25**, 9-26, 9-27
180° Power-off approach	9-25, **9-26**
Power required	4-5, 4-6, 4-15, 4-16, 4-18, 9-11, 13-3, 16-7, 16-11
Preflight	**17-9**
Preflight assessment of the aircraft	**2-1**
Preparation and preflight	**11-7**
Pre-takeoff procedures	2-21, 6-1, 13-12, 13-15
Prior to takeoff	**6-1**
Propellers	3-19, 12-5, 12-7, 12-8, 13-3, 13-5, 13-6, 13-7
Propeller snychronization	**13-7**

R

Recovery from overspeed conditions	5-2
Rectangular course	**7-5**, 7-6, 7-7, 7-9
Rejected takeoff	**13-17**, **16-15**, 16-16
Rejected takeoff/engine failure	**16-14**
Retractable landing gear	2-10, 5-17, 5-24, 8-3, 12-1, **12-12**, 18-4
Controls and position indicator	**12-12**
Emergency gear extension systems	**12-14**
Landing gear safety devices	**12-12**
Landing gear systems	**12-12**
Electrical landing gear retraction system	12-12
Hydraulic landing gear retraction system	12-12
Operational procedures	**12-14**
Approach and landing	**12-16**
Preflight	**12-14**
Takeoff and climb	**12-15**
Reverse thrust and beta range	**15-8**
Risk and resource management	**2-12**
Identifying the hazard	**2-12**
Resource management	**2-13**
Aeronautical decision-making	**2-14**
Flight deck resource management	2-14
Situational awareness	**2-13**
Task management	**2-14**
Risk	**2-13**
Risk assessment	**2-13**
Role of the FAA	**1-2**
Role of the flight instructor	**1-6**

Role of the pilot examiner	1-6
Role of FSTDs and airplanes in UPRT	**5-7**, **5-8**
Rotation and lift-off	**16-16**

S

Safety considerations	**1-10**, **8-5**, 11-10
Secondary stall	5-16, **5-18**, 5-21, 5-25, 5-26
Setting power	**16-2**
Short-field approach and landing	**9-20**, 9-21, **13-22**
Short-field landing	2-22, 9-24, 12-4, 13-22, **14-8**
Short-field takeoff	6-12, 13-17, **14-4**, 17-4
Short-field takeoff and climb	**13-17**
Short-field takeoff and maximum performance climb	**6-8**
Initial Climb	**6-12**
Lift-off	**6-12**
Takeoff roll	**6-11**
Sideslip	5-22, 5-25, 6-6, 6-7, 6-8, 9-12, 9-13, 9-14, 9-15, 9-16, 9-20, 13-21, 13-22, 13-23, 13-25, 13-27, 13-28, 13-29, 13-32, 14-6
Single-engine service ceiling	13-11, 13-36
Slotted flap	12-4
Slow acceleration of the jet engine	**16-3**
Slow final approach	**9-31**
Slow flight	3-6, 5-1, **5-9**, 5-10, 5-11, 5-12, 5-15, 5-27, **13-18**, 17-13
Maneuvering in slow flight	**5-11**
Soft-field approach and landing	**9-23**
Soft-field landing	**14-8**
Soft/rough-field takeoff and climb	**6-13**
Initial climb	**6-13**
Lift-off	**6-13**
Sources of flight training	**1-8**
Airman certification standards (ACS)	**1-9**
Flight safety practices	1-10 1-10
Collision avoidance	**1-10**
Positive transfer of controls	1-10
Runway incursion avoidance	**1-12**
Stall awareness	**1-13**
Use of checklists	**1-15**
Practical test standards (PTS)	**1-9**
Specific excess power (P_S)	4-3, 4-8, 4-14, 4-16, 4-17, 4-18

Speed margins	**16-4**
Speed sense	1-1
Spin awareness	**5-22**, 5-27, **13-18**, 17-4
Spin procedures	**5-23**
Developed phase	**5-24**
Entry phase	**5-24**
Incipient phase	**5-24**
Recovery phase	**5-24**
Spiral dive	5-24, **2-26**, 5-27
Spilt flap	12-4, 17-12
Sport pilot certificate	17-2, **17-3**
Stabilized approach	9-2, **9-4**, 9-5, 9-6, 9-21, 9-22, 9-30, 11-10, 12-4, 13-20, 13-22, 15-13, 16-4, 16-18, 16-21, 16-22
Stall characteristics	**5-15**, 5-18, 5-19, 13-14, 13-19, 16-8
Stall recognition	5-12, **5-13**, 5-22
Angle of attack indicators	**5-13**
Feel	5-13
Hearing	5-13
Kinethesia	5-13
Vision	5-13
Stalls	**5-12**
Accelerated approach to stall	13-20
Engine inoperative loss of directional contol demonstration	13-16
Full stall	5-12, **13-20**
Impending stall	5-12, 5-13, 5-15, 5-16, **5-17**, 5-20, 5-21, 13-18, 13-26, 16-8
Power-off stalls	**5-17**
Power-on stalls (takeoff and departure)	**5-18**
Spin awareness	**5-22**
Stall training	1-7, 5-9, **5-16**, 5-20, **13-18**
Standard traffic patterns	**8-1**, 8-2, 8-5
Base leg	8-3, 8-4
Crosswind leg	8-4
Departure leg	8-4
Downwind leg	8-3, 8-4, 8-5
Entry leg	8-3
Starting, taxiing, and runup	**11-8**
Steep spiral	4-17, **10-3**, 10-4
Steep turns	3-12, 3-14, 3-16, 5-6, 5-19, **10-1**, 10-2, 10-3, 11-7, 17-4, 17-13, 18-17

Straight-and-level flight	3-1, 3-3, **3-6,** 3-7, 3-8, 3-16, 3-19, 3-20, 3-25, 7-2, 7-13, 7-16, 10-4, 10-5, 11-7, 13-19, 18-18, 18-20, 18-21
Level flight	**3-8**
Straight flight	**3-6**
S-turns across a road	**7-8**, 7-9, 7-10
System malfunctions	9-29, 18-10, **18-13**
Electical system	**18-13**, 18-14
Pitot-static system	**18-14**

T

Takeoff and climb	5-18, 6-1, 6-2, 6-6, 6-11, 6-13, **11-8**, 12-4, **12-16**, 13-10, 13-12, **13-15**, **13-17**, 13-33, 15-2, **16-12**, 16-14, 16-16, 16-17, 17-4, **17-12**,
Takeoff and departure	6-1, **13-9**, 15-12, , 16-14
Takeoff checks	**2-22**
Takeoff roll	2-22, 6-1, **6-3**, 6-5, **6-6**, 6-7, 6-10, **6-11**, **6-13**, 6-14, 12-6, 13-14, 13-15, 13-30, **14-3**, 14-4, 16-2, **16-12**, 16-14, 16-16
Ground roll	2-20, 3-1, 6-1, 6-3, 6-11, 6-14, 9-10, 13-15, 13-35, 14-4, 4-7, 18-5
Takeoffs	5-9, 5-15, 6-1, 6-5, 6-6, 6-8, 6-10, 6-12, 6-13, 8-1, 9-15, 9-18, 11-1, 11-6, 11-7, 11-8, 12-4, 12-9, 13-15, 13-17, 14-1, 14-3, 14-4, 14-8, 15-12, 15-13, 16-14, 16-15, 16-20,
Taxi and Taxiing	1-13, 2-14, 2-15, **2-18**, 2-19, 2-20, 2-21, 2-22, 6-1, 6-6, 6-10, 9-9, 9-36, 11-5 , **11-8**, 13-14, 13-15, 13-21, 13-36, 14-1, **14-2**, 14-3, 14-4, 14-8, 15-9, 16-1, 16-12, 16-24, **17-12**
Terrain selection	18-1, **18-4**
Terrain types	**18-5**
Confined areas	**18-5**
Trees (forest)	**18-5**, 18-6
Water (ditching) and snow	**18-6**
Thrust reversers	**16-10**, 16-11, 16-19, 16-24
Thrust to thrust lever relationship	**16-2**
Timing	1-1, 7-1, 7-7, 7-10, 7-18, 9-15, 9-32, 14-6, 16-16, 18-4
Tires	2-**10**, 2-11, 2-22, 2-23, 9-18, 9-23, 9-33, 9-35, 9-37, 14-4, 16-15, 16-16, 16-24, 17-12,
Total mechanical energy	4-1, 4-2, 4-3, 4-4, 4-16, 4-18, 4-19
Touchdown	**9-9**, **9-17**, **9-33**, **9-37**, **9-38**, **14-4**, **16-24**
Crosswinds	**9-15**, 9-16, **9-17**, **9-18**, 9-19, 9-20, 9-33, 13-22, 14-2, 14-4, **14-6**, 17-6, 18-7
Three-point landing	14-4, **14-5**, 14-6
Wheel landing	14-4, **14-5**, 14-6
Touchdown in a drift or crab	**9-35**
Tracking over and parallel to a straight line	**7-5**
Training considerations	**13-35**, **15-13**, 15-15
Flight training	13-35, 13-36, 15-14, 15-15

Ground training	**15-14**
Training for night flight	**11-7**
Transition training	**12-17**, 12-18, 14-15, 15-13, 17-4, 17-7, 17-13, 17-15
Transition training considerations	**17-4**
Flight instructors	**17-4**
Flight school	**17-4**
Trim control	3-5, **3-10**, 17-12
Turbine inlet temperature (TIT)	12-10, 12-11
Turbocharging	**12-9**, 12-10, 12-11
Ground boosting versus altitude turbocharging	**12-10**
Heat management	**12-10**, 12-11
Operating characteristics	**12-10**
Turbocharger failure	**12-11**
Low manifold pressure	**12-11**
Over-boost condition	**12-11**
Turboprop airplane electrical systems	**15-10**, 15-11
Turboprop engines	1-23, 13-9, **15-2**
Truboprop engine types	**15-2**
Fixed shaft	**15-2**, 15-3, 15-4, 15-5, 15-9, 15-11
Split shaft/free turbine engine	**15-5**, 15-6, 15-7, 15-9, 15-11, 15-13,
Turbulent air approach and landing	**9-20**

U

Unusual attitude versus upsets	**5-1**
Upset prevention and recovery	**5-1**, 5-5, 5-27
Upset prevention and recovery training (UPRT)	**5-4**, 5-27
Use of power	**9-31**, 10-3

V

V_1	16-12, 16-14, 16-15, 16-16
Maximum V_1	16-15
Minimum V_1	16-15
Reduced V_1	16-15, 16-16
Variation of thrust with rpm	**16-3**
Visibility	2-4, 3-17, 7-1, 8-4, 8-5, 8-6, 9-2, 11-3, 11-4, 11-6, 11-9, 11-14, 13-16, 13-20, **14-3**, 14-8, 16-24, 17-12, 18-7, 18-22
Visual inspection of the aircraft	2-2
Visual preflight assessment	2-1, **2-4**
V-speeds	13-1, 13-2, **16-12**
V_{LOF}	13-10, 13-15, 16-12, 16-16

V_{MC}	13-2, 13-15, 13-17, 13-18, 13-20, **13-23**, 13-24, 13-25, **13-26**, 13-27, 13-28, 13-29, 13-31, 13-35, 13-37, 16-12
V_R	2-22, 13-1, 13-10, 13-14, 13-15, 16-12, 16-14, 16-16
V_{REF}	13-1, 16-20, 16-22
V_{SSE}	13-1, 13-18, 13-25, 13-26, 13-35, 15-36,
V_X	3-17, 4-17, 6-10, 6-11, 6-12, 6-13, 13-1, 13-17, 13-19, 13-22,
V_{XSE}	13-1, 13-17,
V_Y	3-17, 4-8, 4-17, 5-26, 6-4, 6-5, 6-6, 6-10, 6-11, 6-12, 6-13, 13-1, 13-15, 13-16, 13-17, 13-19, 13-22, 13-31,
V_{YSE}	13-1, 13-12, 13-16, 13-18, 13-20, 13-26, 13-27, 13-29, 13-30, 13-31, 13, 32, 13-34, 13-35, 13-37

W

Weather considerations	17-4, **17-7**
Weathervaning	2-19, 2-20, 6-8, 9-18, 9-36, 14-1, **14-2**, 14-3, 14-7, 14-8,
Weight and balance	2-4, 5-23, 5-26, 13-11, **13-12**, 13-13, 17-9
Basic empty weight	13-12, 13-13, 13-14
Empty weight	13-12, 13-13,
Maximum landing weight	13-14
Ramp weight	13-14
Standard empty weight	13-12, 13-13,
Zero fuel weight	13-13
Weight and balance requirement related to spins	**5-26**
Wheelbarrowing	6-12, 9-20, **9-35**, 9-37, 13-17
Wing rising after touchdown	**9-37**

Y

Yaw damper	**13-8**, 13-16
Yaw string	13-27, 13-29